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PERSONAL PORTFOLIO KNOWLEDGE BASE
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ABOUT

Name:
Sonal Patani

Professional Profile:

Sonal Patani is an Information Technology student currently pursuing Master of Science in Information Technology (M.Sc IT). She has completed Bachelor of Science in Information Technology (B.Sc IT).

She is passionate about solving real-world problems using technology. Her main areas of interest include Artificial Intelligence, Machine Learning, Data Science, Cloud Computing, Internet of Things, and Full Stack Web Development.

She has developed multiple academic and personal projects involving web applications, database management, IoT automation, and AI-based solutions.

Her goal is to continuously improve technical knowledge, build industry-level projects, and contribute to innovative software solutions.

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EDUCATION DETAILS
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Master of Science in Information Technology (M.Sc IT)
Status: Currently Pursuing

During M.Sc IT, she gained knowledge in:
- Advanced Programming Concepts
- Artificial Intelligence
- Machine Learning
- Data Mining
- Database Management System
- Cloud Computing
- Software Engineering
- Computer Networks
- Operating System Concepts

Bachelor of Science in Information Technology (B.Sc IT)
Status: Completed

Key learning areas:
- Programming fundamentals
- Web technologies
- Database systems
- Computer fundamentals
- Software development concepts

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TECHNICAL SKILLS
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Programming Languages:

Python:
Experience with:
- Data handling
- Automation scripts
- Machine learning implementation
- AI applications
- Backend logic development

PHP:
Experience with:
- Server side scripting
- Dynamic websites
- Database connectivity
- CRUD operations

JavaScript:
Experience with:
- Interactive websites
- Client side programming
- Frontend functionality

Frontend Development:

HTML:

- Website structure creation
- Semantic web pages
- Forms and layouts

CSS:
- Responsive design
- Styling
- UI improvement
- Better user experience

JavaScript:
- DOM manipulation
- Website interactions

Database Skills:

MySQL:
Knowledge of:
- Database creation
- Tables
- Relationships
- SQL queries
- Joins
- CRUD operations
- Data management

Cloud Computing:

AWS Cloud:
Knowledge of:
- EC2 instances
- Cloud server basics
- Application deployment
- Virtual machines

Developer Tools:

Git:
- Version control

- Code management

GitHub:
- Repository creation
- Project documentation
- Collaboration

VS Code:
- Code development
- Debugging

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PROJECT 1:
ELEGANCE INTERIOR DESIGN WEBSITE
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Project Type:
Full Stack Web Development

Problem Statement:

Many interior design businesses need an online platform where customers can explore services and previous design work.

Traditional offline methods make it difficult to reach more customers and manage information.

Solution:

Elegance Interior Design Website provides an online platform where users can view interior design services and administrators can easily manage website content.

Main Modules:

User Module:

Users can:
- Visit website
- View interior services
- Explore designs
- Check company information
- Contact business

Admin Module:

Admin can:
- Login securely
- Add new records
- Update existing information
- Delete unnecessary data
- Manage website content

Features:

1. Authentication System
Secure admin login system.

2. Content Management
Admin can dynamically manage information.

3. Database Integration
All records are stored and managed using MySQL.

4. CRUD Operations
Implemented:
Create
Read
Update
Delete

Technology Used:
- HTML
- CSS
- JavaScript
- PHP
- MySQL

Learning Outcome:

This project improved knowledge of:
- Full stack development
- Database connectivity
- Backend programming

- Admin panel development

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PROJECT 2:
IoT SMART ROOM LIGHT SYSTEM
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Project Domain:
Internet of Things

Problem Statement:

Manual light control increases inconvenience and unnecessary energy usage when people forget to switch lights on or off.

Solution:

An IoT based smart lighting system that automatically controls room lights based on sensor input.

Components Used:

ESP Module:

Works as the main controller of the system.

PIR Sensor:

Purpose:
Human motion detection

Working:
Detects movement inside room and sends signal to controller.

LDR Sensor:

Purpose:
Light intensity detection

Working:
Measures surrounding brightness level.

System Workflow:

Step 1:
Sensors collect environment data.

Step 2:
ESP module processes sensor values.

Step 3:
Decision is made based on conditions.

Step 4:
Light automatically turns ON or OFF.

Features:

- Automatic room light control
- Motion based detection
- Brightness detection
- Smart automation

Technology Used:
- ESP Module
- Arduino IDE
- Embedded C
- Sensors

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PROJECT 3:
AI RESUME SCREENING SYSTEM
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Project Domain:
Artificial Intelligence

Problem Statement:

Companies receive hundreds of resumes. Manual resume checking requires more time and effort.

Solution:

AI based resume screening system automatically analyzes resumes and matches candidates with job requirements.

Features:

Resume Upload:

Candidate uploads resume file.

Text Extraction:

System extracts important information.

Data Processing:

Extracted information is cleaned and processed.

Skill Matching:

Candidate skills are compared with required skills.

Ranking:

System generates candidate score.

Technology Used:

- Python
- Machine Learning
- Natural Language Processing
- Streamlit

Concepts Used:

NLP:
For understanding resume text.

Machine Learning:
For intelligent decision making.

Data Processing:

For cleaning and preparing data.

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PROJECT 4:
RAG BASED AI DOCUMENT CHATBOT
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Full Form:
Retrieval Augmented Generation

Problem Statement:

Traditional chatbots cannot answer questions from personal documents.

Solution:

A RAG chatbot allows users to upload documents and ask questions based on uploaded knowledge.

Architecture:

Document Input:

Users upload:
- PDF files
- Text documents

Text Extraction:

System reads document data.

Chunking:

Large text is divided into smaller meaningful parts.

Embedding Generation:

Text chunks are converted into numerical vectors.

Vector Database:

Embeddings are stored for fast searching.

Retrieval:

Relevant information is searched based on user questions.

Generation:

AI model creates accurate answers using retrieved information.

Technology Used:

Python:
Main programming language.

LangChain:
For building RAG pipeline.

FAISS / ChromaDB:
For vector storage.

LLM:
For answer generation.

Streamlit:
For chatbot interface.

Advantages:

- Reduces hallucination
- Gives document-based answers
- Fast searching
- Personalized AI assistant

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ACHIEVEMENTS
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- Qualified GATE 2026 Computer Science / IT
- Completed B.Sc Information Technology
- Developed multiple technical projects

- Experience in AI, Cloud, IoT, and Web Development

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CAREER OBJECTIVE
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To work as a technology professional where I can apply my skills in software development, artificial intelligence, data science, and cloud computing to solve real-world problems.

Interested Roles:

- AI Developer Intern
- Machine Learning Intern
- Data Science Intern
- Software Developer Intern
- Web Developer Intern
- Cloud Engineer Intern

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CHATBOT SAMPLE QUESTIONS
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Question:
Tell me about Sonal.

Question:
What technical skills does Sonal have?

Question:
Explain Elegance Interior Design project.

Question:
What is the IoT Smart Light System?

Question:
Explain Sonal's AI projects.

Question:
Why should we select Sonal for internship?

Question:
What technologies has Sonal worked with?

Question:
What are Sonal's career goals?

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END OF KNOWLEDGE BASE
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